Evaluating Explanation Methods for the Classification of Land Use and Land Cover

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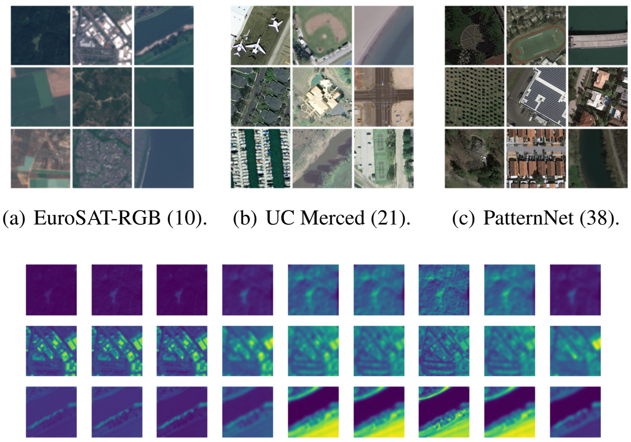
Abstract. We discuss the land use and land cover classification problem for remote sensing (RS), training each of three deep learning architectures on four commonly studied datasets, including on multi-spectral data. While achieving performance on par with the state of the art, we discuss the need for interpretable outputs and hence apply three popular Explainable AI (xAI) techniques, including SHAP and class activation map (CAM) variants, to each of our twelve trained models. Directly addressing the lack of quantitative evaluation of xAI techniques within the RS domain, and the deep learning field more generally, for each xAI method we investigate four properties of the Co-12 framework proposed by Nauta et al. [20] in Correctness, Output-Completeness, Contrastivity, and Compactness via six unique metrics, some for the first time to our knowledge in the RS domain. We find that there are no shortcuts for explainability, with some explanations that look convincing, particularly via CAM methods, in fact highly misrepresentative of actual model behaviour, especially for land cover images without a clearly defined object. We also highlight how the performance of xAI methods can vary wildly between dataset and model combinations, underscoring the lack of a 'one size fits all' approach to explainability and the need, even beyond RS, to not just study how we develop increasingly complex models, but how we build explanations to interpret them too.



(a) EuroSAT-RGB (10). (b) UC Merced (21). (c) PatternNet (38).

(d) EuroSAT-MS (10) versions of the top 3 images from (a), normalised band/channel-wise (i.e. by column). From left to right, the bands are "B02", "B03", "B04", "B05", "B06", "B07", "B08", "B8A", "B12".

Figure 1. Image samples from each of the four RS datasets considered in this work. Total number of classes for each dataset in (brackets).

1 Introduction

Satellites orbiting the earth beam back terabytes of data daily among which lie incredibly valuable insights that already inform many areas of policy and decision making on domestic, national as well as international levels in areas across agriculture, natural hazard tracking, urban planning, and monitoring the impacts of climate change. However, this satellite and aerial image data (together constituting remote sensing (RS) data), and in particular the task of Land Use and Land Cover (LULC) classification which will constitute the focus for the remainder of this work, presents a unique set of challenges compared to typical imaging data. Firstly, the earth's geographically diverse surface results in images that rarely conform to a single object distribution, instead possessing a complex multi-object structure. Not only is there this "coexistence" [2] of objects within the same image, but these objects can appear at hugely varying scales, contributing to the problem of *intra*-class diversity. Shared features between images, the notion of *inter*-class similarity, compound this issue. Some land cover images may not even contain a primary 'object'. Secondly, the multi-spectral (MS) nature of raw satellite data vastly increases the quantity and complexity of data involved and processing required.

Deep learning methods, in particular Convolutional Neural Networks (CNNs), have been unquestionably successful in processing these data but their fundamental 'black box' mode of operation presents a problem to policy makers and a highly-scientific field which is historically accustomed to a more physically grounded approach. This explainability problem therefore continues to restrict models' use [36], as well as the development of new techniques. Explainable AI (xAI) methods attempt to address this issue by surfacing insights into the underlying decision-making process of complex models through a variety of approaches, including by approximating model behaviour locally, e.g. LIME [24], or by back-propagating network activations, e.g. DeepLIFT [30] and the commonly visualised Class Activation Maps (CAMs) [39]. But there is a documented lack of quantitative evaluation of these xAI techniques [11, 20], with authors often relying solely on anecdotal evidence and cherry-picked visual examples to supposedly demonstrate the efficacy of their selected methods. This has made it incredibly difficult to concretely compare approaches and to identify aspects of the explanations which need to be improved, hindering their development and general adoption. While some authors do evaluate their methods directly via user studies, these results are still difficult to compare and replicate, while also being very time and resource intensive. This motivated Nauta et al. [20] to propose a metrics framework in the form of their "Co-12 properties", providing a comprehensive set of evaluation criteria that can be applied to all xAI methods.

In this work therefore, we investigate four properties of this framework (Correctness, Output-Completeness, Contrastivity, and Compactness) via six quantitative metrics to evaluate three distinct xAI methods (PartitionSHAP [16], Grad-CAM [28], and KPCA-CAM [13]) on the LULC problem in the RS domain, some for the first time to the best of our knowledge. Specifically, we apply and evaluate these xAI methods on three foundational deep learning architectures in ResNet-50 [8], ConvNeXt [15], and the Swin Transformer (SwinT) [14]. Each of these models is trained to performance on par with the state of the art on four well established LULC datasets: EuroSAT [9] (in both its -RGB and -MS variants) as an example of multi-spectral data designed to maximise intra-class variation and inter-class similarity so as to discourage a learned dependence on low-level features; UC Merced (UCM) [38] as a seminal high-resolution RS dataset; and PatternNet [40] for investigating methods' success for a larger number of classes. Example images from each dataset are shown in Figure 1.

In summary, our work thereby offers the following contributions:

1. An end-to-end, easily extensible and modular framework (with full source code) for evaluating the explanations of deep learning models following the Co-12 properties categorisation.
2. Qualitative and quantitative evaluation results of this framework applied to RS data, notably including MS data.
3. An examination of the generalisation ability of the selected xAI methods to modern deep learning architectures outside their original scope and their flexibility to the RS domain.
4. A discussion of the application suitability of existing xAI evaluation metrics to RS images.

2 Related Work

Höhl et al. [11] provide a thorough review of existing applications of xAI to the RS domain, most importantly highlighting the variety of purposes for which reliable explanations of model outputs are desired within the field: *to control* models' behaviour, for example to adapt to practical out of domain contexts; *to discover* previously unknown characteristics or biases in the input data, thereby informing future feature extraction and deep learning methods; *to improve* models' overall performance by providing a further avenue for evaluation and to validate that expected image regions form the basis for outputs; and *to justify* outputs in decision-critical contexts requiring accountability, such as the allocation of agricultural subsidies.

An additional challenge Höhl et al. [11] discuss is that xAI techniques are often developed for classical computer vision datasets, e.g. ImageNet [26], and so their relevance or 'fit' to a specific field such as RS is hard to quantify. While there have been some efforts to develop RS-specific xAI variants such as CSG-CAM proposed by Guo et al. [7], applications of xAI to the LULC problem [29, 31, 33] rarely evaluate the efficacy of their approaches instead assuming the correctness of the selected method and proceeding to qualitatively apply their results to diagnose model performance via a determination of transferable semantics, globally important bands, and incorrectly learned proxies respectively.

Similarly, metrics by which to evaluate xAI methods have been primarily focused and tested on ImageNet-like general object detection tasks such as in [1, 25]. However, Tomsett et al. [32] highlight that the lack of consistent application of even these general metrics in the literature leads to inconsistencies and ultimately makes the results unreliable. They therefore strongly advocate authors to avoid relying on just one or two metrics to measure a property and to only compare metrics in an implementation-aware fashion, in doing so underscoring the need for transparent metric frameworks and configurations.

Most similar to our work, Kakogeorgiou and Karantzalos [12] evaluate xAI methods on RS data but ultimately only quantitatively examine one metric for each of three of the Co-12 properties CORRECTNESS, CONTINUITY, and COMPACTNESS (property classification ours). While they do consider a greater number of distinct xAI methods (nine), they omit the most popular xAI methods in the RS domain specifically in SHAP and CAM-variants beyond the original Grad-CAM [11]. Throughout their discussions, they also focus primarily on generated explanations' adherence to domain expectations in terms of identifying the anticipated regions of interest, which Nauta et al. [20] explicitly caution against. The lack of accompanying source code or full configuration parameters also hinders the usefulness of their results for the reasons discussed above.

3 Methodology

This section provides an overview of the selected xAI methods in Section 3.1 before moving on to detail the quantitative metrics we use to evaluate them in Section 3.2.

3.1 xAI methods

We examine three different *post-hoc* (explaining an already trained predictive model) xAI methods, all of which generate heatmaps of individual pixel importance given a model and an input image to be explained. A high-level summary of these methods is provided in Table 1.

Table 1. High-level summary of considered xAI methods

xAI method	Key idea	Required input
SHAP [16]	Game theoretic Shapley values	Model output on perturbed inputs
Grad-CAM [28]	Weighted summation of feature maps	Output class gradients wrt feature maps
KPCA-CAM [13]	First kernel principle component	Feature map activations

3.1.1 SHAP

SHapley Additive exPlanations (SHAP) [16] attribute an importance value ϕ_q (the game theoretic Shapley value) to each 'feature' (an individual image pixel), x_q, representing the positive or negative contribution of the inclusion of that feature to the final output/class confidence, f (also referred to in the game theory literature as the "payout"). These ϕ_q are formally defined in terms of feature index subsets $S \subseteq F$ as:

$$\phi_q = \sum_{S \subseteq F \setminus \{q\}} \frac{|S|!(|F| - |S| - 1)!}{F!} \left[f(\boldsymbol{x}_{S \cup \{q\}}) - f(\boldsymbol{x}_S) \right]. \quad (1)$$

Calculating this exactly requires summing over all possible feature coalitions (subsets S) with exponential runtime, 2^Q (where $Q = hw$ is the number of pixels in an image of height h and width w), so practical use (especially for images with high numbers of features/pixels) requires the use of approximating techniques (see Section 4.1).

3.1.2 Grad-CAM

Gradient Class Activation Maps (Grad-CAM) [28] are a fast way (requiring only one forward and backward pass through the model being explained) to directly generate explanations based on weighted network activations. For the given score (the output logit before softmax is applied) of a particular classification class c, y^c, Grad-CAM weights each feature map, $A^k \in \mathbb{R}^{u \times v}$, of the selected convolution-like layer by a factor α_k^c in a linear combination, retaining only positive influences on the final output class by applying the ReLU function ($\mathrm{ReLU}(x) = \max(0, x)$):

$$L^c_{\textit{Grad-CAM}} = \mathrm{ReLU}\left(\sum_{k=1}^{n} \alpha_k^c A^k\right) . \tag{2}$$

These "neuron importance weights", α_k^c, are designed to capture the importance of feature map k to the classification y^c via a partial linearisation. They are therefore calculated via a global average pool (that is, over all units, i, j, of each feature map) of the gradients (calculated via backpropagation) of y^c with respect to each feature map unit A^k_{ij} , as given by:

$$\alpha_k^c = \frac{1}{uv} \sum_{i=1}^{u} \sum_{j=1}^{v} \frac{\partial y^c}{\partial A^k_{ij}} . \tag{3}$$

Since $L_{\text{Grad-CAM}} \in \mathbb{R}^{u \times v}$, for practical use and interpretation it is upscaled to match the dimensions of the input image, $\boldsymbol{x} \in \mathbb{R}^{h \times w}$.

3.1.3 KPCA-CAM

A much more recent variant of CAM designed to capture non-linear relationships within CNN activations more effectively, KPCA-CAM's authors claim it produces more precise activation maps [13]. KPCA refers to Kernel Principal Component Analysis, an extension of principle component analysis (PCA) which first applies a *non-linear* mapping of the input data into a higher dimensional space through the use of a kernel function before applying PCA. The kernel function the original authors show to be most effective is the sigmoid kernel, defined between two $\mathbb{R}^n$ feature vectors, $\boldsymbol{x}_a$ and $\boldsymbol{x}_b$, as:

$$k(\boldsymbol{x}_a, \boldsymbol{x}_b) = \tanh(\gamma\, \boldsymbol{x}_a^\top \boldsymbol{x}_b + c_0) , \tag{4}$$

where γ is the scale/slope parameter (commonly set to $1/n$, as in `scikit-learn`) and c_0 is the intercept/offset (set to 1 by default).

For the case of explaining an input image via KPCA-CAM, there are $u \cdot v$ feature vectors (one per unit of the feature maps) each consisting of n activation values, one for each feature map in the layer. Formally this is given by $\boldsymbol{x}_{(i-1)u+j} = (A^1_{ij}, \cdots, A^k_{ij}, \cdots, A^n_{ij})^\top$, using the same notation as in eq. (3). To produce the final CAM, $L_{\textit{KPCA-CAM}}$, we project the kernel matrix $K \in \mathbb{R}^{uv \times uv}$, defined via $K_{ab} = k(\boldsymbol{x}_a, \boldsymbol{x}_b)$, onto its largest (that is, corresponding to the largest eigenvalue) eigenvector, $\boldsymbol{v}_1 \in \mathbb{R}^{uv}$:

$$L_{\textit{KPCA-CAM}} = \underset{\mathbb{R}^{uv} \to \mathbb{R}^{u \times v}}{\mathrm{reshape}} (K\boldsymbol{v}_1) . \tag{5}$$

This equates to each feature vector, $\boldsymbol{x}_c$, being projected onto the first (since we take only the largest eigenvector) principal component. Note that this technique does not require the model's gradients, merely the activations (feature maps) of the selected convolution-like layer from one forward pass on the input image. Therefore, it can only be used to explain the model's actual output (unlike Grad-CAM which works for any desired class c). The same upscaling process as for $L_{\text{Grad-CAM}}$ is still needed.

3.2 Quantitative evaluation framework

In this section, we discuss the metrics we use to evaluate the effectiveness of each xAI method. Their precise Python implementation can be viewed in the accompanying source code. We follow the "Co-12 properties" categorisation framework [20] and adopt the same notation where f denotes the predictive model being explained and e denotes an explanation method which generates explanations such that $g(\boldsymbol{x}) = e(f(\boldsymbol{x}))$ ultimately represents the explanation generated by e for the output of f evaluated on input sample $\boldsymbol{x}$.

3.2.1 Ranked explanations

In most cases, metrics require the consideration of 'important' pixels: these are determined by converting explanation heatmaps for the output class of f into 'ranked' explanations assigning a value of 0 to the pixel in the explanation with the highest value and a value of $Q - 1$ to the pixel with the lowest value (potentially negative). This can be formalised as redefining the output of e, the heatmap matrix H with value $H_{ij} = H(i,j) = H(\boldsymbol{r}_p)$ at pixel coordinates $\boldsymbol{r}_p = (i,j)$ in the image, as an ordered (ranked) set, $\mathcal{R}$:

$$\begin{gathered}\mathcal{R} = (\boldsymbol{r}_0, \cdots, \boldsymbol{r}_p, \cdots, \boldsymbol{r}_{Q-1}) \\ \text{where } p_1 < p_2 \Leftrightarrow H(\boldsymbol{r}_{p_1}) > H(\boldsymbol{r}_{p_2}) \text{ as in [27].}\end{gathered} \tag{6}$$

3.2.2 Explanation similarity

Many metrics also necessitate the calculation of a notion of similarity between generated explanations. To enable comparison between xAI methods which use different output ranges, we only ever compare ranked explanations. Following Adebayo et al. [1], we use the perceptual structural similarity index measure (SSIM) [35], which captures structural information more effectively than a more rudimentary mean squared error (MSE) comparison, where a score of 1 indicates the greatest similarity.

Since we are often only interested in the areas deemed most important by e, we also calculate the top-m intersection, I_m, suggested by Ghorbani et al. [5], between two ranked explanations $\mathcal{R}_1$ and $\mathcal{R}_2$ as:

$$I_m(\mathcal{R}_1, \mathcal{R}_2) = \frac{|(\mathcal{R}_1)_{0:m-1} \cap (\mathcal{R}_2)_{0:m-1}|}{m} , \tag{7}$$

where a score of 1 again indicates the highest possible similarity.

Finally, the ranked nature of explanations naturally lends itself [1, 5] to the calculation of the Spearman rank-order correlation coefficient, ρ, between two explanations, where a score of 0 indicates no correlation (completely dissimilar) and ± 1 indicates positive/negative correlation (exactly the same/inverses of each other).

3.2.3 Evaluating metrics

We first examine the property of CORRECTNESS which addresses the "descriptive accuracy" [19] of e to f; in other words, e's faithfulness to the true inner workings of the black box f. It is crucial to note that an explanation which looks plausible to a user may not be true to f's reasoning. In the worst case, explanations could merely be acting as edge detectors on the input, simply reflecting the underlying model architecture as demonstrated by Adebayo et al. [1]. As a basic 'sanity check' therefore, we perform a **Model Parameter (Cascading) Randomisation Test** [1], fully randomising all model parameters and calculating the similarity, as described in Section 3.2.2, between the explanations generated for this randomised model and those of the trained model for the same inputs. For a faithful e sensitive to f, we expect a low similarity.

This is a necessary but not sufficient condition on the correctness of e, so we also calculate the **Area Under Most Relevant First**

Figure 2. The effect of various deletion methods on the same randomly selected 60% pixels of a `wastewater treatment plant` image from PatternNet. From left to right: the original image, constant replacement, Gaussian noise, blurring, in-painting, fill via nearest neighbour, and pixel shuffling.

Incremental Deletion Curve (AUC-MoRF) metric as proposed by [23, 27], and similarly to the work by Kakogeorgiou and Karantzalos [12]. For each iteration k of K total iterations, we 'delete' the top $t_k = \lfloor hw \cdot k/K \rfloor$ most relevant pixels from $\boldsymbol{x}$ (that is, the pixels at coordinates $\boldsymbol{r}_p \in \mathcal{R}$ such that $p < t_k$) to produce a new input $\boldsymbol{x}'(\mathcal{R}, k)$ and record the model's confidence in the original image's predicted class but for the new input, $f_c(\boldsymbol{x}')$. Notably, there are several methods of deleting pixels/generating $\boldsymbol{x}'$, visualised in Figure 2.[2] We can then plot the (discrete) *incremental deletion curve*, $f_c(\boldsymbol{x}'(\mathcal{R}, k))$, over k, as visualised in Figure 3(d), and calculate the area under this curve (AUC) using the trapezoidal rule:

$$\mathrm{AUC}(f, \boldsymbol{x}, \mathcal{R}) = \frac{1}{2} \sum_{k=1}^{K} \left[f_c\left(\boldsymbol{x}'(\mathcal{R}, k-1)\right) + f_c\left(\boldsymbol{x}'(\mathcal{R}, k)\right) \right] \, . \tag{8}$$

Given the importance to view this AUC only in the context of a baseline [12, 20], we compute the ratio of this area to that of the randomised incremental deletion curve which uses the mean value of $f_c(\boldsymbol{x}'(\tilde{\mathcal{R}}, k))$ for an ensemble of T randomised ranked explanations, $\tilde{\mathcal{R}}_t$,[3] informing deletion:

$$\mathrm{AUC}_{ratio} = \frac{\mathrm{AUC}(f, \boldsymbol{x}, \mathcal{R})}{\frac{1}{T}\sum_{t=1}^{T} \mathrm{AUC}(f, \boldsymbol{x}, \tilde{\mathcal{R}}_t)} \, . \tag{9}$$

For a good explanation, we expect $\mathrm{AUC}_{ratio} < 1$ since this indicates that e was better than random at identifying pixels truly important to f with a more steeply decreasing deletion curve and hence a lower AUC.

Related to this metric as being a point on the incremental deletion curve, we also evaluate OUTPUT-COMPLETENESS (OC), which assesses how much of f's behaviour e is able to describe, via the **Deletion and Preservation Check** metrics in the vein of Wagner et al. [34]. For the deletion check, this is simply calculated as the change in confidence, $\Delta\mathcal{C}$, versus the original input at a particular point on the deletion curve: $\Delta\mathcal{C}_{del} = f_c(\boldsymbol{x}'(\mathcal{R}, \mu K)) - f_c(\boldsymbol{x})$ for some proportion $\mu \in (0, 1)$. For the preservation check, we invert $\mathcal{R} \to \mathcal{R}^{-1}$ (to delete the pixels deemed *least* important first) and then calculate $\Delta\mathcal{C}_{pres} = f_c(\boldsymbol{x}'(\mathcal{R}^{-1}, (1-\mu)K)) - f_c(\boldsymbol{x})$ instead. Again, these values should only be viewed in comparison to their randomised versions, $\Delta\tilde{\mathcal{C}}_{del/pres}$, as was done for the AUC metric:

$$\mathrm{OC}_{del} = \Delta\tilde{\mathcal{C}}_{del} - \Delta\mathcal{C}_{del} \tag{10}$$

$$= \frac{1}{T}\sum_{t=1}^{T} \left[f_c(\boldsymbol{x}'(\tilde{\mathcal{R}}_t, \mu K)) \right] - f_c(\boldsymbol{x}'(\mathcal{R}, \mu K)) \, , \tag{11}$$

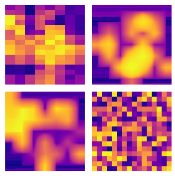
$$\mathrm{OC}_{pres} = \Delta\mathcal{C}_{pres} - \Delta\tilde{\mathcal{C}}_{pres} \, . \tag{12}$$

The latter is defined so that the best score for both metrics is 1 (that is, when $\Delta\mathcal{C}_{del} \approx -1$ as we deleted important pixels and $\Delta\tilde{\mathcal{C}}_{del} \approx 0$; and when $\Delta\mathcal{C}_{pres} \approx 0$ as we kept important pixels and $\Delta\tilde{\mathcal{C}}_{pres} \approx -1$) with negative values indicating randomly selecting pixels for deletion/preservation was more effective at decreasing/preserving model confidence.

If f's output changes significantly, even changing its predicted class, for a perturbed input $\tilde{\boldsymbol{x}}$, we desire an equally significant change in g. Miller [17] asserts that, in line with the "contrastive explanations" given by humans when asked to explain an event, e should not just explain why the output was f but should also explain why the output was not f'. We therefore evaluate CONTRASTIVITY as a measure of the discriminativeness of e with respect to f's output, the aim being to enable comparison to other outputs [20]. In a very similar fashion to Nie et al. [22], we probe the **Target Sensitivity** of our selected xAI methods by generating an adversarial example [6] of $\boldsymbol{x}$, $\tilde{\boldsymbol{x}}$ (a virtually imperceptibly perturbed input designed to fool f to produce an entirely different output), and comparing the explanations for the outputs $f(\boldsymbol{x})$ and $f(\tilde{\boldsymbol{x}})$, g and $\tilde{g}$ respectively. We seek a *large* difference (low similarity) between g and $\tilde{g}$ for a good explanation model e.

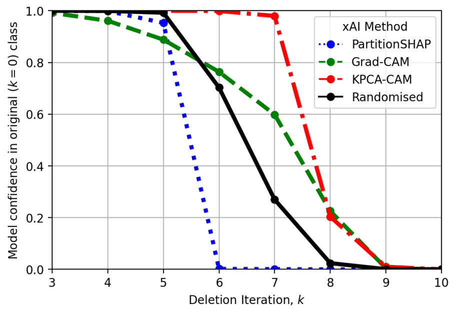
Finally, we propose to measure the COMPACTNESS of generated explanations directly (i.e. not via ranked versions) via a sim-

(a) `PermanentCrop` image taken from EuroSAT-RGB.



(b) Explanations left to right, top to bottom: PartitionSHAP, Grad-CAM, KPCA-CAM, randomised.

(c) Partially deleted images (using `shuffle`) at $k = 6$ ($K = 14$).



(d) Incremental deletion curve, $f_c(\boldsymbol{x}'(\mathcal{R}, k))$.

Figure 3. The process of evaluating the CORRECTNESS of our selected xAI methods (compared with randomised explanation) via the incremental deletion curve, described in Section 3.2.3. A smaller area under the curve (AUC) is better. PartitionSHAP has an AUC of 5.45 compared to the AUC of a randomised explanation of 6.49. Grad-CAM and KPCA-CAM are both worse than random for this example.

[2] A common criticism of pixel perturbation/deletion methods is that they can lead to samples outside of f's input domain, as observed by Hooker et al. [10]. The shape of the deleted region has also been shown to potentially leak class information to the model [25]. However, further analysis of deletions in RS images specifically is beyond the scope of this work.

[3] In practice, we generate random ranked explanations, $\tilde{\mathcal{R}}_t$ in the form of a $d \times d$ square grid with each grid element of uniform importance to ensure that nearby pixels are assigned similar rankings, just as they are in non-randomised ones, $\mathcal{R}$. In Figure 2, we use $d = 8$ for example.

ple **Sparsity**, S, calculation in terms of g's normalised heatmap, $\overline{H}_{ij} = \text{abs}(H_{ij})/\max_{k,l}(H_{kl})$, parametrised by some threshold, $\tau \in (0, 1)$:

$$S(H, \tau) = \frac{\left|\{H_{ij} : \overline{H}_{ij} \leq \tau\}\right|}{|H|}, \tag{13}$$

where $|X|$ denotes the number of elements of a matrix or set, X. S thereby represents the proportion of pixels *not* marked as important (i.e. those having low normalised heatmap values, $\overline{H}_{ij}$), calibrated via τ. Explanations should be sparse (high S) with a minimal number of important pixels to avoid overwhelming users: less is more.

4 Results

4.1 Experimental configuration

We train our twelve (4 datasets $\times$ 3 models) *{dataset, model} combinations* (DMCs) using `pytorch`.[4] To maximise usage of available GPU memory, we use a batch size of 64/48/32 when training the ResNet/ConvNeXt/SwinT models respectively. For EuroSAT-MS, we first select a nine-band subset (see Figure 1(d)) of the full thirteen bands available, given that bands "B01", "B09", "B10", and "B11" are collected mainly for the purposes of atmospheric corrections and preprocessing (e.g. filtering out high cloud-cover images, which has already been performed by the creators of EuroSAT) and not LULC observations, similarly to Engelbrecht and van Zyl [4]. Since our selected datasets have well balanced classes (a similar number of samples per class), we simply report the overall accuracy (accuracy $= \frac{\text{correct predictions}}{\text{total predictions}}$) of our models, achieving performance on par with the state of the art as shown in Table 2.

Table 2. Trained model accuracy

Dataset	Model[5]	Proven accuracy by [3][6] or by given citation (%)	Our accuracy (%)
UCM	ResNet-50	98.57	97.77
	ConvNeXt	98.26 [37]	98.84
	SwinT	98.57	97.77
EuroSAT-RGB	ResNet-50	98.83	99.00
	ConvNeXt	98.78	98.34
	SwinT	98.94	98.80
EuroSAT-MS	ResNet-50	96.43 [31]	96.73
	ConvNeXt	*No existing work*	97.75
	SwinT	*to our knowledge.*	97.34
PatternNet	ResNet-50	99.74	99.52
	ConvNeXt	99.67	99.08
	SwinT	99.69	99.59

Having fully trained all models, for each in turn we then proceed to generate explanations using the SHAP, Grad-CAM, and KPCA-CAM xAI methods for 32 randomly selected samples from each class of each test dataset. We use the PartitionSHAP method from the `shap` package[7] to approximate SHAP with features, x_q, removed via blurring and `max_evals=3000`. For both CAM-based methods, we utilise the implementation by the `pytorch-grad-cam`,[8] selecting the final convolutional layer (with n output channels/feature maps) of each model, believed to capture the highest-level concepts (with the largest receptive field) required for ultimate classification while still retaining the spatial information lost in later linear layers. For SwinT, we extract the values after the final normalisation layer and before average pooling,[9] reshaping them to the same dimensions ($n \times u \times v$) as a typical CNN convolution layer as expected by the CAM methods.

We then evaluate the six metrics described in Section 3.2.3 for these explanations, following the metric configuration detailed in Table 3. For all datasets apart from EuroSAT-MS, we adopt the pixel shuffling deletion method (randomly rearranging pixels within the desired region) as this is empirically observed to strike a good balance between allowing random perturbations to affect model output and qualitatively maintaining overall image structure to attempt to avoid creating out of domain samples as discussed in footnote 2. For EuroSAT-MS, all three models are incredibly sensitive to shuffling-based deletions[10] resulting in evaluation metrics showing little difference between explanation-informed and randomised approaches, with both slashing model confidence even when just a small proportion of pixels are affected. To effectively evaluate the xAI methods on this dataset therefore, we use the blurring deletion method instead. To generate adversarial examples for assessing target sensitivity, we use the DeepFool (ℓ_∞) method [18].

Table 3. Evaluation metric configuration

Property/Metric description	Configuration
Top-m intersection similarity	$m = \lfloor hw/10 \rfloor$
Random explanation generation grid size	$d = 16$
Random explanation, $\tilde{\mathcal{R}}_t$, ensemble size	$T = 5$
Incremental deletion iterations for CORRECTNESS	$K = 14$
Deletion/Preservation proportion for OUTPUT-COMPLETENESS	$\mu = 0.2$
Sparsity threshold for COMPACTNESS	$\tau = 0.5$

4.2 Metric-based evaluation of explanations

It should first be noted, as by Nauta et al. [20], that it is unlikely that a particular explanation method scores highly across every single property/metric—some metrics are also partially mutually exclusive—so it is best to view explanations holistically, selecting an explanation which meets the needs of a particular use case. That being said, we proceed to examine the properties in turn. A highly generalised summary of our evaluation results, presenting the mean for each item across all twelve combinations, can be found in Table 4 and selected data are visualised in more detail via box plots in Figure 4. Full data, up to the level of individual class labels, are provided in `xlsx` and `DataFrame` form in the accompanying source code.

The model randomisation test acts primarily as a sanity check for CORRECTNESS so 'optimal' values vary based on the nature of the explanation itself. That being said, the explanations produced by KPCA-CAM of the trained model are significantly more similar to those of the randomised model by the Spearman's rank and top-m intersection metrics when compared to the other two methods. This is particularly pronounced for the {EuroSAT-MS, ResNet-50} DMC,

[4] We perform all our training and experiments using version 2.5.1 and on a 10GB MIG partition of a NVIDIA A100 GPU. We use the 60/20/20% train/validation/test dataset splits of Neumann et al. [21] as provided by `torchgeo`, except for PatternNet where we use our own randomised split.

[5] We use the `Small` variant of both the ConvNeXt and SwinT models.

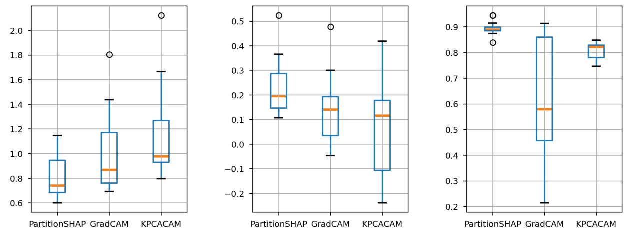
[6] Dimitrovski et al. [3] use ConvNeXt-Tiny, the variant one step smaller than the Small variant we use with approximately 20 million fewer parameters.

[7] PartitionSHAP documentation: https://bit.ly/4jyVOnp.

[8] Package repository: https://github.com/jacobgil/pytorch-grad-cam.

[9] For the `torchvision` implementation we use, this corresponds to the output of the `Permute` module at https://bit.ly/43bZypc.

[10] We theorise this is because MS data are more often discontinuous (see Figure 1(d)) and so shuffling leads much more quickly to out of domain samples. The lower comparative resolution of some bands (even after upscaling) potentially exacerbates this issue.



(a) AUC_{ratio} ($\downarrow$) (b) OC_{pres} ($\uparrow$) (c) Sparsity, S ($\uparrow$)

Figure 4. Box plots of selected metrics, each considering 12 DMC data points per xAI method. The central orange line is the median and the box extends to the first and third quartiles. Whiskers extend to the furthest point within 1.5$\times$ the interquartile range. Flier points are circles past the whiskers.

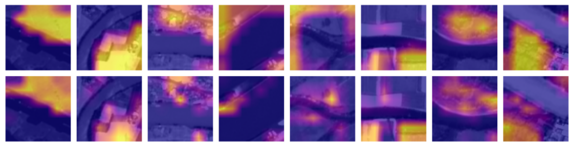
with explanations highlighting near identical regions (even if the intensity is weaker), as for the `River` class in Figure 5(a).

For the AUC_{ratio} metric, plotted in Figure 4(a) and exemplified step by step in Figure 3, only PartitionSHAP performs better than random on average. The CAM methods only outperform randomised explanations in approximately half of DMCs and in some cases perform far worse, with the 1.8+ fliers for the CAM methods both corresponding to the {PatternNet, ResNet-50} DMC.

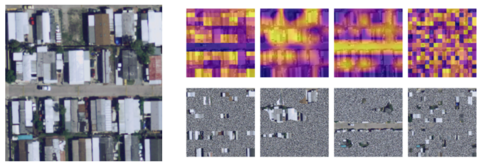
PartitionSHAP is also significantly more successful in identifying important pixels when measured by metrics assessing OUTPUT-COMPLETENESS and is always (apart from for the {PatternNet, ResNet-50} DMC) at least better than random ($\text{OC}_{pres} > 0$). Again, KPCA-CAM often actually performs *worse* than a random selection of important pixels, especially for SwinT DMCs. It is also surprisingly inconsistent between DMCs, as is clear from the whisker extent in Figure 4(b). A comparison of the ability of different explanation methods to effectively preserve relevant pixels is given in Figure 5(b).

In terms of CONTRASTIVITY, both Grad-CAM and PartitionSHAP's explanations reassuringly change markedly from their original form for adversarial inputs across all three similarity metrics. Explanations generated via KPCA-CAM however, change far less than we would expect, with on average 62.7% of pixels shared between considered explanations and a very high average SSIM value too (0.627): KPCA-CAM does not appear to be very class sensitive, producing similar explanations for visually similar inputs even if the model output has changed completely.

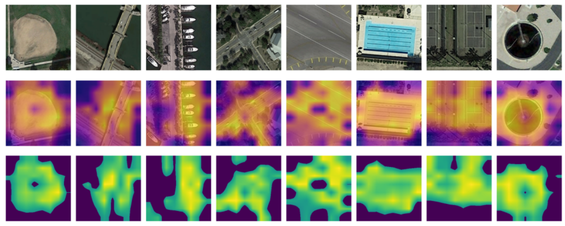
Finally examining COMPACTNESS, both PartitionSHAP and KPCA-CAM consistently produce very compact explanations across DMCs, highlighting only approximately up to 10-25% of image pixels as important as can be seen plotted in Figure 4(c). However, Grad-CAM demonstrates huge variability in its COMPACTNESS with some explanations almost uniform ($S = 0.2$-0.3) for SwinT DMCs as exemplified in Figure 5(c) – such explanations provide little actionable insight into a model's decision-making process. Grad-CAM's explanations for ResNet-50 DMCs also have lower S around 0.5-0.6.



(a) KPCA-CAM explanations for a trained {EuroSAT-MS, ResNet-50} DMC (top) are often very similar to a randomised model (bottom).



(b) Preservation check images (bottom) to measure OUTPUT-COMPLETENESS for a `mobilehomepark` sample (left) from the {UCM, SwinT} DMC for explanations from PartitionSHAP, Grad-CAM, KPCA-CAM, and randomised (top). The confidence in the original output class drops from $1.0 \to \{0.9864, 0.0102, 0.0005, 0.7030\}$ (a smaller drop is better).



(c) Grad-CAM explanations (middle) have poor COMPACTNESS for the {PatternNet, SwinT} DMC. The bottom row visualises the normalised explanation heatmaps, $\overline{H}$, (with $\overline{H}_{ij} < \tau$ set to 0) where $S = \{0.41, 0.44, 0.46, 0.46, 0.25, 0.34, 0.29, 0.40\}$.

Figure 5. *All descriptions correspond to the images from left to right.* Visualisation and quantitatively-informed evaluation of selected explanations. In the explanations, yellow, lighter regions indicate important pixels; purple, darker regions unimportant pixels.

5 Discussions

5.1 Observations

Across all metrics, KPCA-CAM performs poorly, with a consistent pattern of explanations which do not reflect underlying model behaviour. This is especially evident in explanations' poor CONTRASTIVITY. This could very likely be due to the lack of gradient information in its formulation inhibiting its class discriminative ability, which the original authors also observed in their comparisons with other CAM methods [13]. Explanations frequently merely look like edge detectors or colour segmentation maps, especially unhelpful for RS images which may lack a clearly defined object. The originally proposed evaluations on the ImageNet dataset [26] which showed "more precise activation maps" may simply have been a product of more clearly defined single object images rather than being reflective

Table 4. Summary of Evaluation Results (Mean)

xAI Method	COR – RT[a] ρ	COR – RT[a] I_m	COR – RT[a] SSIM	COR – AUC_{ratio} ($\downarrow$)	OC_{del} ($\uparrow$)	OC_{pres} ($\uparrow$)	CON – TS ($\downarrow$)[b] ρ	CON – TS ($\downarrow$)[b] I_m	CON – TS ($\downarrow$)[b] SSIM	COM – S ($\uparrow$)[c]
PartitionSHAP	0.027	0.146	0.423	**0.813**	**0.113**	**0.233**	**0.090**	0.198	**0.450**	**0.891**
Grad-CAM	0.062	0.120	0.343	1.074	0.013	0.145	0.315	**0.194**	0.506	0.590
KPCA-CAM	0.129	0.186	0.346	1.240	−0.028	0.047	0.641	0.627	0.627	0.797

Best in **bold**. [a] CORRECTNESS – Randomisation Test [b] CONTRASTIVITY – Target Sensitivity [c] COMPACTNESS – Sparsity of the method's capability to explain the underlying models; that is to say, the intersection over union (IoU) metric was indicative of good object localisation but not necessarily faithful explanations.

Separately, while they may lack flexibility to the RS domain specifically, across all metrics no methods show significantly worse performance on ConvNeXt or SwinT (both developed after PartitionSHAP and Grad-CAM were originally proposed) in comparison with ResNet-50. This suggests all three explanation methods can be reasonably applied to new design paradigms and are flexible to new building block combinations. Reassuringly, neither does the deepness of these two newer networks appear to affect explanation quality. Considered conversely however, the poor relative performance on ResNet-50 is also surprising, especially for Grad-CAM. The only exception is in Grad-CAM's even lower COMPACTNESS scores for SwinT DMCs, likely again a result of the evaluation domain with large areas of similar token activity within the transformer a consequence of classification by texture/region-information rather than via the localised feature information considered by the other two more traditional convolution-based architectures. PartitionSHAP seems to avoid these issues due to its model-agnostic consideration of every pixel region separately rather than as part of a singular feature map from within the model itself.

It should also be noted that none of the explanation methods separately considered the rich multi-channel nature of the input to the MS models, relying only on images' spatial information rather than this dimensional aspect. Empirical evaluations [9, 31] have concretely demonstrated the varying importance of different bands and their combinations to deep models, all of which are not accounted for here. Likewise, the simplification of incredibly complex models to a heatmap representing a single layer is potentially hugely reductive and it may be more insightful into full model behaviour to find a way to visualise relations between layers themselves, building up to ultimate classification. This is especially the case for CAM methods, such as those considered, which are originally of a much lower resolution and are scaled up for visualisation.

5.2 Limitations of evaluation metrics

The primary concern is as discussed in footnote 2 and exemplified in Figure 5(b): pixel deletion methods can quickly lead to out of domain samples, potentially invalidating the use of the original model. It is possible PartitionSHAP only scored highly in metrics utilising such deletion because its important pixels were often more evenly distributed throughout the image and therefore less likely to change the prediction class, rather than because their significance to the model was actually higher than those proposed by the CAM methods. This could explain why all models performed best on SwinT for the AUC_{ratio} metric since the transformer image token-based structure may be more robust to deletion perturbations localised to just a few 8×8 pixel windows but not to ones distributed throughout the image affecting many image tokens, as in the random explanations.

This work also does not thoroughly investigate the impact of differing methods of generating such random explanations. The grid based solution used in this work does not produce 'realistic' explanations – an approach based on assigning super-pixel regions random importance values could have provided a better evaluation baseline.

Considering COMPACTNESS specifically, it could be argued that explanations for LULC images should *not* be unconditionally compact at all, given there may not be a distinct object in the scene and a general type of land cover to consider instead. However, this also raises the argument that existing explanation methods focused on saliency heatmaps highlighting particular regions may not be suitable to LULC problems in the first place.

6 Conclusions

Taken together, our evaluation results and discussions show that there are no shortcuts to reliable explainability: the notion of a 'good' explanation and associated xAI method is a complex, multi-faceted concept which is *not* domain-agnostic and depends on the specific data and model combination being considered. However, our metrics do show that some existing xAI methods, notably KPCA-CAM, can produce deceptive explanations, at least when applied outside of their original target domain, emphasising that care should be taken to select xAI methods for each particular use case and tailored to the needs of end users. We therefore recommend that authors probe the target sensitivity of their explanations thoroughly to ensure that a model *decision* and not just an image itself is being explained. It also highlights that examining just a few properties, only COHERENCE and CORRECTNESS in the case of KPCA-CAM [13] (metric classification ours), is insufficient to confidently establish the efficacy of such methods.

Equally, our results have shown that popular methods in the domain, in Grad-CAM and SHAP, do perform well but that their application should still be justified and tuned for the task under consideration with an awareness that they can struggle with localising important regions for LULC imagery, resulting in uniform, uninformative explanations.

Reaffirming the findings of [11, 20], we have demonstrated that a quantitative assessment of explanation quality is possible and provided an extensible framework by which such evaluations can be more easily undertaken by future work. However, following our application of these metrics to the LULC problem, we have also discussed their flaws, thereby highlighting the need for domain-specific adaptions to be developed and the demand for standardisation within domains to enable robust comparison of future proposed xAI methods. We believe that such standardisation of parameters (e.g. τ for COMPACTNESS) should ideally be determined via user-studies to ensure explanations can meet the specific needs and expectations of domain end users.

6.1 Future work

As touched on several times by this work, a stand-alone investigation of the impact of using different deletion methods for the evaluation of explanations specifically in the RS domain is still needed, especially for MS data for which models appear to be incredibly sensitive to perturbations. Methods to adequately provide explanations for more complex models utilising band combinations should also be explored by future work. While we have focused on largely model-agnostic xAI methods applicable to both transformers and CNNs, the recent explosion in transformer-based computer vision architectures in particular brings with it a need for architecture-specific explanations such as visualising attention weights alongside appropriate new metrics which can be used to evaluate their effectiveness. Apart from also considering further properties not evaluated in this work, including COHERENCE and CONTINUITY (utilising datasets providing segmentation maps for instance), future work could additionally investigate the extent to which explanations and their faithfulness to model behaviour differ for *incorrect* model outputs specifically, a distinction not made by this work.

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